

A Data Science Approach for Evaluating Experience Variables in Insurance Risk Modeling

Naibo Hu
Independent Researcher
Chicago, IL, USA
naibo.hu@outlook.com

Abstract—Experience variables are fundamental in insurance pricing for incorporating historical loss information. However, their effectiveness in detecting underlying risk structures is not always well understood. This paper presents a practical framework for constructing experience-based features and investigates their role in detecting risk in insurance portfolios through a controlled simulation study. Three experimental scenarios are designed to represent different risk structures: random risk, persistent latent risk, and risk fully explained by observable rating variables. Generalized linear models (GLMs) with Tweedie distributions are used to evaluate predictive performance across these scenarios. The results show that experience variables provide substantial predictive improvement when persistent latent risk exists but offer limited benefit in other settings. These findings highlight the role of experience-based features as proxies for persistent latent risk in insurance portfolios.

Index Terms—insurance pricing, experience variables, risk modeling, simulation study, generalized linear models

I. INTRODUCTION

Experience-based variables have long been used in the insurance industry to incorporate historical claims information for assessing risk and expected loss. Early actuarial research studied this concept within the framework of experience rating and credibility theory, where past claims are used to estimate the underlying risk associated with a policyholder [1]. Subsequent developments extended classical credibility to incorporate varying exposure structures, improving the reliability of historical claims information used in credibility estimation [9].

In modern insurance analytics, experience-based variables are commonly incorporated into regression-based statistical pricing models such as generalized linear models (GLMs) [2], [10]. They remain widely used across different lines of business, in contrast to many observable rating variables that are line-specific. For example, automobile insurance models rely on driver and vehicle characteristics, while commercial liability and property models use payroll, building characteristics, and catastrophe exposure. These observable predictors tend to be line-specific, whereas experience variables provide a common mechanism for incorporating historical claims information across different insurance products.

Despite their widespread use, the mechanism through which experience variables improve predictive performance remains insufficiently understood. In particular, it is unclear whether these variables capture persistent latent risk within an insurance portfolio or simply reflect random variation in observed

claims. If experience variables capture latent risk, they provide insight into underlying policyholder heterogeneity that is not directly observable through standard rating variables. Otherwise, their predictive contribution may be limited. Understanding when experience variables provide genuine predictive value is therefore an important question for both actuarial pricing practice and modern predictive modeling.

This paper first describes how experience variables are constructed in practical insurance pricing models. Building on this formulation, we develop a controlled simulation framework to examine how experience variables perform under different underlying risk structures. Three scenarios are considered: random risk, persistent latent risk, and risk fully explained by observable rating variables. Using Tweedie generalized linear models (GLMs), we evaluate how experience variables affect predictive performance across these risk environments. The findings provide insight into when experience variables capture meaningful information about latent risk and when their predictive contribution is limited.

II. CONSTRUCTION OF EXPERIENCE VARIABLES

The construction of an experience variable begins with the aggregation of historical claims information for a given risk entity. Historical experience is typically measured relative to exposure, where exposure represents insured risk over a given period, typically expressed in policy-years or exposure units. In practice, this experience is captured through metrics such as claim frequency, loss ratios, or pure premiums accumulated over multiple historical periods. Let R_i^{act} denote the historical rate for policy i , defined as the ratio of losses to the corresponding exposure.

Next, credibility adjustments are applied to combine the historical rate R_i^{act} with the corresponding portfolio-level expected rate R_i^{exp} [3]. The expected rate represents a baseline risk estimate obtained from portfolio-level benchmarks or manual rates that reflect the average claims experience for similar risks within the industry. In this framework, the credibility weight Z_i determines the relative weight assigned to the historical rate and the expected rate, placing greater weight on R_i^{act} as more exposure becomes available.

In practical implementations, credibility is often modeled as an increasing function of exposure. A commonly used specification can be expressed as

$$Z_i = \min \left(1, \left(\frac{V_i}{\tau} \right)^\gamma \right) \quad (1)$$

where V_i represents the exposure associated with policy i , and τ is a threshold parameter controlling the exposure level at which credibility approaches one. As an illustration, in a commercial auto portfolio where exposure is measured by the number of insured vehicles, a threshold such as $\tau = 100$ vehicles implies that policies with fewer vehicles receive partial credibility, while larger fleets approach full credibility. The parameter γ controls the smoothness of the credibility function. Smaller values of γ result in slower growth of credibility with exposure, placing greater weight on the portfolio benchmark when exposure is limited, while larger values allow individual experience to influence the estimate more rapidly.

The final calculated experience variable can be expressed as

$$EV_i = \frac{Z_i R_i^{act} + (1 - Z_i) R_i^{exp}}{R_i^{exp}} \quad (2)$$

where EV_i represents the experience variable used in the GLM pricing model and is normalized by the expected rate. The resulting value is a multiplicative relativity centered around one. Values greater than one indicate unfavorable historical experience, meaning the policy has generated more claims or losses than expected relative to the portfolio benchmark, while values below one indicates more favorable experience. In practical pricing models, this relativity serves as an adjustment to the expected risk level: policies with $EV_i > 1$ receive a debit adjustment, whereas policies with $EV_i < 1$ receive a credit adjustment.

This construction reflects actuarial practice, where credibility-weighted loss experience relative to portfolio expectations is incorporated into generalized linear pricing models, providing a structured mechanism for integrating claims history into risk prediction.

III. SIMULATION FRAMEWORK

A. Policy Portfolio Generation

The simulation contains 30,000 policies observed annually from 2012 through 2024, resulting in 390,000 policy-term observations in total. The first five policy years (2012–2016) are used to construct five-year rolling historical experience variables. Therefore, 2017 is the first policy year for which a complete experience history is available.

Each policy is assigned a seven-digit policy number. For every policy year, an effective date is randomly generated within the corresponding calendar year, and the policy year is defined as the coverage year derived from the effective date. A policy-term identifier is constructed by concatenating the policy number and the effective date so that each policy-term observation is uniquely identified.

Several observable characteristics are simulated for each policy. The average driver age is randomly drawn from a uniform distribution between 18 and 70 years, while the

average vehicle age is drawn from a uniform distribution between 0 and 15 years. Fleet size, representing the number of insured vehicles, is generated from a right-skewed exponential distribution and capped at 300 vehicles. This design reflects commercial auto portfolios, where a single policy may insure multiple vehicles. As policies progress through time, driver age and vehicle age increase gradually each year, while fleet size may fluctuate slightly across policy terms to reflect changes in insured exposure.

In addition, the number of traffic violations for each policy term is simulated using a Poisson distribution, which is commonly used to model the number of discrete events occurring over a fixed exposure period [4]. The expected violation rate depends on driver age, with younger drivers assigned higher violation frequencies to reflect commonly observed patterns in insurance portfolios. The expected violation count is also scaled by the number of insured vehicles, so that policies with larger fleets have proportionally more opportunities for violations to occur.

B. Claim Portfolio Generation

Claims are simulated separately at the claim level and stored in the claim table before being aggregated to the policy-term level. Claim generation begins by determining the expected claim frequency for each policy term. A baseline claim frequency of 0.04 is specified in the simulation. The expected claim frequency is then calculated as the product of the baseline frequency, a driver age factor, a vehicle age factor, and a violation factor derived from the number of violations per insured vehicle. Specifically, the adjusted claim frequency is defined as

$$f_i = f_0 \times a_i \times v_i \times w_i \quad (3)$$

where f_0 denotes the baseline claim frequency, a_i represents the driver age factor, v_i represents the vehicle age factor, and w_i represents the violation factor for policy term i .

The driver age factor is defined as 1.6 when the average driver age is below 25, and 1.0 otherwise, reflecting the higher claim frequency typically observed among younger drivers. The vehicle age factor is calculated as $1 + \frac{\text{vehicle_age}}{40}$, allowing claim frequency to increase gradually with vehicle age. The violation factor is defined as $1 + \text{violations_per_vehicle}$, where $\text{violations_per_vehicle}$ represents the number of violations divided by the number of insured vehicles. The resulting adjusted frequency represents the expected claim rate per insured vehicle for the policy term. The expected number of claims is obtained by multiplying the adjusted frequency by the number of insured vehicles. This step ensures that policies insuring more vehicles have proportionally more opportunities for claims to occur. The actual number of claims for each policy term is then simulated using a Poisson distribution, with the mean equal to the expected number of claims. [4].

Next, if the simulated number of claims for a policy term is greater than zero, a severity amount is generated independently

for each claim. Claim severities are simulated using a lognormal distribution, as insurance loss amounts are typically right-skewed with a small number of large claims [5]. The mean severity is defined as the product of a baseline severity and a severity adjustment factor derived from observable characteristics. In the simulation, the baseline severity is set to 5,000, and the severity adjustment factor depends on driver age, vehicle age, and violation history. Specifically, the severity adjustment factor depends on three components: driver age, vehicle age, and violation history. Younger drivers are assigned a higher severity multiplier, vehicle age increases severity gradually through a scaling factor, and higher violations per vehicle introduce an additional multiplicative factor. Once the mean severity is determined, individual claim amounts are drawn from the lognormal distribution. To prevent extreme losses from dominating the simulated portfolio, each simulated claim amount is capped at 120,000 before being recorded as the ultimate incurred amount and stored in the claim table. Since a single policy term may produce multiple claims, the claim table is aggregated to the policy-term level to compute the total incurred loss and claim counts for each policy term.

After claim losses are generated, a manual expected loss (MEL) is calculated for each policy term. In insurance ratemaking, the manual expected loss provides a portfolio-level benchmark for expected losses and is commonly used as a reference point when evaluating policy performance [3]. In the simulation, the MEL is calculated as the product of the number of insured vehicles, the baseline claim frequency (0.04), the baseline claim severity (5,000), and a pricing margin factor (1.3). Therefore, the expected loss per insured vehicle equals $0.04 \times 5,000 \times 1.3 = 260$, and the final MEL for a policy term is obtained by multiplying this value by the number of insured vehicles.

C. Model Target

The modeling target is constructed to measure the deviation between realized losses and the expected loss. In the simulation, the target variable for policy term i is defined as

$$\text{Target}_i = \frac{\text{Trended Ultimate Incurred Loss}_i}{\text{MEL}_i}. \quad (4)$$

This formulation normalizes realized losses relative to the expected benchmark and places policies with different exposure levels on a comparable scale. To obtain the trended ultimate loss, simulated claim amounts are first adjusted using a loss development factor (LDF) and then a trend factor. In insurance ratemaking, LDFs are used to project reported losses to their ultimate expected level, reflecting the fact that claim payments and reserve estimates develop over time as claims mature [6]. Trend adjustments are applied to account for changes such as inflation or shifts in cost levels over time. Both adjustments are standard practices in actuarial pricing to ensure that losses are evaluated on a consistent cost level. In the simulation, the LDF is set to 1.1 and the trend factor is set to 1.05. The resulting value represents the trended ultimate incurred loss used in constructing the modeling target.

D. Experience Variable

For each policy term, historical claim information from the five preceding policy years is used to construct the experience variable. Following the formulation introduced in Section II, the historical rate R_i^{act} is defined as the five-year historical claim frequency, calculated as the total number of claims observed during the five-year window divided by the corresponding cumulative exposure, where exposure is measured as the total number of insured vehicles accumulated over the same period. The expected rate R_i^{exp} represents the portfolio-level benchmark frequency, which is set to 0.04.

The credibility weight Z_i is determined based on cumulative exposure using a credibility threshold of 200 vehicles in the simulation. Policies whose cumulative exposure exceeds this threshold are assigned full credibility, allowing the historical claim frequency to dominate the credibility-weighted estimate. Policies with smaller exposure levels receive partial credibility, placing greater weight on the portfolio-level expected frequency. The parameter γ in the credibility function is set to 0.3, which results in a gradual increase in credibility as exposure grows. Larger values of γ would cause credibility to increase more rapidly with exposure, while smaller values lead to a slower progression toward full credibility.

The final experience variable EV_i is computed according to Equation (2), combining the credibility-weighted historical claim frequency and the expected portfolio frequency. The resulting value represents a multiplicative relativity centered at one.

IV. EXPERIMENTAL DESIGN

The goal of the experiment is to evaluate the predictive power of experience variables under different underlying risk structures. To achieve this, three risk scenarios are simulated: *random risk*, *latent persistent risk*, and *fully observable risk*.

A. Risk Scenarios

1) *Random Risk*: Random risk refers to a setting in which policy risk does not contain a persistent underlying component over time. Instead, claims are driven primarily by random fluctuation. In the simulation, this scenario is implemented by determining claim frequency using only observable policy characteristics. In addition, small random variations are introduced to these characteristics across policy years so that risk conditions fluctuate over time rather than remaining persistent. As a result, claim losses for a policy are largely driven by short-term variation instead of a stable underlying risk level.

2) *Latent Persistent Risk*: Latent persistent risk refers to a setting in which each policy is associated with an unobserved risk characteristic that remains stable over time. This latent trait is hidden and systematically embedded in past claims. In the simulation, a policy-level latent trait is generated for each policy by drawing a random value from a normal distribution with mean 0 and standard deviation 0.4. This value remains fixed across all policy years for that policy, ensuring that the same underlying risk level persists over time. As a result, policies with higher latent traits consistently produce higher

claim activity across multiple years, while policies with lower traits exhibit lower claim activity. This latent trait is introduced to mimic persistent unobserved risk differences across policies. For example, two policies with identical observable characteristics may still exhibit different claim patterns over time. Under this setting, past claim experience provides information about future risk.

3) *Fully Observable Risk*: Fully observable risk refers to a setting in which the underlying claim risk is largely explained by observable rating variables. As a result, policies with similar observable characteristics are expected to exhibit similar claim behavior. In the simulation, the influence of observable variables is strengthened in the claim generation process. Specifically, stronger multipliers are applied to observable characteristics so that claim frequency and severity depend more heavily on driver age, vehicle age, and violations per vehicle. Under this specification, most of the systematic variation in claim outcomes is explained directly by observable characteristics.

These three scenarios are designed to evaluate the predictive value of experience variables under different underlying risk structures. In the random risk scenario, claim variation is primarily driven by random fluctuation, so historical experience is not expected to provide substantial predictive improvement. In the latent persistent risk scenario, hidden policy-level risk remains stable over time, and the experience variable is expected to provide meaningful predictive value. In the fully observable risk scenario, claim risk is largely explained by observable rating variables, so the incremental contribution of the experience variable is expected to be minimal. Comparing model performance across these scenarios allows us to assess when experience variables improve predictive performance and when their benefit is limited.

B. Modeling Framework

The modeling dataset covers policy years 2017–2024. Data from 2017–2021 are used for model training to fit the Tweedie GLM, while data from 2022–2024 are reserved for validation. This design ensures that model performance is evaluated on unseen data, providing an unbiased assessment of predictive accuracy and preventing information leakage.

Model estimation is conducted using a Tweedie generalized linear model (GLM) with a log link. The Tweedie distribution is widely used in insurance applications because it accommodates outcomes with both zero and positive continuous values and performs well for highly skewed loss distributions [7]. Let Y_i denote the target variable for policy term i with expected value $\mu_i = E[Y_i]$. The model relates the expected loss to the predictors as

$$\log(\mu_i) = \mathbf{x}_i^\top \boldsymbol{\beta}. \quad (5)$$

For each risk scenario, three models are estimated: a basic model using observable rating variables, an experience-only model using the experience variable, and a combined model including both. Comparing these models isolates the contribution

of the experience variable. Table I summarizes the variables used in the modeling framework.

TABLE I
MODEL VARIABLES

Variable Name	Feature Group
Policy Year	Basic
Number of Vehicles	Basic
Average Driver Age	Basic
Average Vehicle Age	Basic
Number of Violations	Basic
Five-Year Credibility Weighted Claim Frequency	Experience
Loss Cost Relativity	Target

C. Summary Statistics

This section summarizes key characteristics of the simulated dataset using the training sample from 2017 to 2021. Table II shows that approximately 8%–9% of policy terms reach full credibility under the exposure threshold, while the majority receive partial credibility. This indicates that, for most policies, the experience variable reflects a combination of historical claims experience and the baseline expected frequency, a pattern commonly observed in real-world insurance portfolios.

TABLE II
DISTRIBUTION OF CREDIBILITY LEVELS FOR THE EXPERIENCE VARIABLE (2017–2021)

Credibility Level	Random Risk	Latent Risk	Observable Risk
Full Credibility (%)	8.76%	8.72%	8.66%
Partial Credibility (%)	91.24%	91.28%	91.34%
Total Policy Terms	150,000	150,000	150,000

Figure 1 illustrates that the distribution of the five-year claim frequency is right-skewed across the three simulated risk scenarios. Although the overall shapes of the distributions are similar, the spread of the experience variable differs across scenarios, as expected from the simulation design. The latent persistent risk scenario produces a wider distribution due to stable unobserved heterogeneity across policies, while the random risk scenario is more concentrated because claim outcomes are primarily driven by stochastic variation. The observable risk scenario shows moderate spread, reflecting stronger effects from observable rating variables. These patterns confirm that the simulation framework generates distinct risk environments while maintaining comparable ranges for the experience variable.

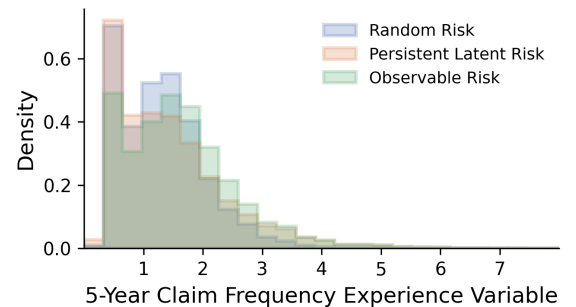


Fig. 1. Distribution of the experience variable (2017–2021).

V. MODEL RESULTS

A. Beta Coefficient Analysis

Across all Tweedie GLM models, the coefficients for the predictors are statistically significant with p-values below 0.05. This indicates that the simulated rating variables contain meaningful information for explaining variation in the target loss cost relativity. Figure 2 summarizes the coefficient of the experience variable across the three risk scenarios under the experience-only model and the combined basic + experience model.

In the random risk scenario, the coefficient of the experience variable is close to zero in both models. When estimated alone, the coefficient is slightly higher because the experience variable absorbs some variation associated with other explanatory variables. However, once observable rating variables are included, the coefficient decreases to nearly zero. This suggests that the experience variable provides minimal incremental predictive information in this setting and has limited ability to capture purely random variation in claim outcomes.

In the latent persistent risk scenario, the results differ substantially. The experience variable exhibits a strong and stable coefficient in both models. In the experience-only model, the coefficient reaches approximately 0.388, indicating a strong relationship between historical claims experience and the target loss cost relativity. When observable variables are included, the coefficient declines only slightly, suggesting that the predictive contribution of the experience variable remains largely unchanged. This indicates that the experience variable captures persistent unobserved risk characteristics embedded in historical claims that are not explained by observable rating variables.

In the fully observable risk scenario, the experience variable shows a moderate coefficient when estimated alone, with a value around 0.132. However, once observable rating variables are included in the model, the coefficient declines to approximately zero. This indicates that the predictive information contained in historical claims experience largely overlaps with the information captured by observable variables. When these signals are explicitly modeled, the incremental contribution of the experience variable becomes negligible.

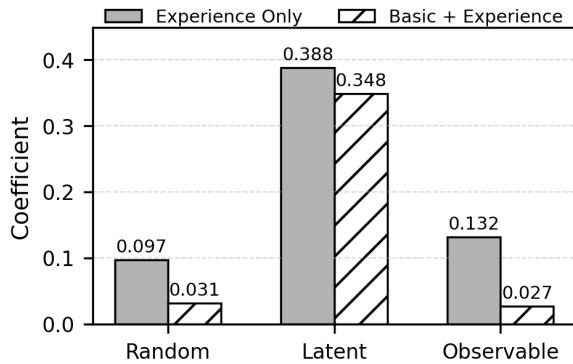


Fig. 2. Beta coefficient of the experience variable across risk scenarios.

While coefficient estimates describe the statistical relationships captured by the model, predictive performance is evaluated using out-of-sample validation metrics such as the Gini coefficient and lift charts.

B. Predictive Performance Analysis

Predictive performance is evaluated using two complementary metrics: the Gini coefficient and lift charts. The Gini coefficient measures the discriminatory power of a model by evaluating how effectively policies are ranked according to risk. Derived from the Lorenz curve, it compares predicted outcomes with actual outcomes to assess how effectively a model ranks policies according to their risk levels [8]. A higher Gini coefficient indicates better separation between high-risk and low-risk policies.

Lift charts provide a complementary view of model performance by comparing predicted and observed values across ordered risk segments. Policies in the validation dataset are ranked according to their predicted loss cost relativities and grouped into bins with approximately equal levels of manual expected loss (MEL). For each bin, the average predicted and actual loss cost relativities are compared. The lift score is calculated as the difference between the lowest- and highest-risk bins. A higher lift score indicates stronger risk separation and therefore better predictive performance.

Figures 3 and 4 summarize model performance across the three simulated risk scenarios using these two metrics.

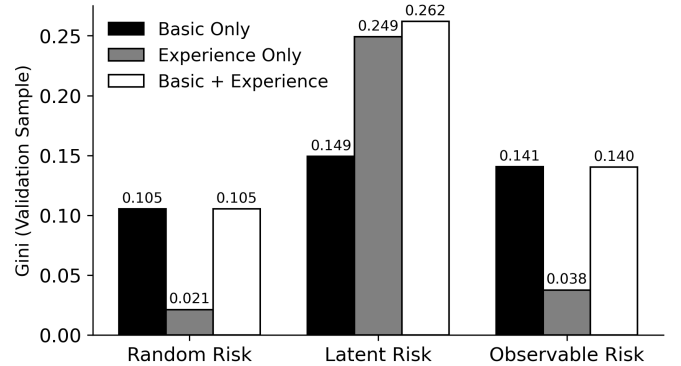


Fig. 3. Comparison of Gini coefficients across the three risk scenarios.

In the random risk scenario, the basic model and the combined basic + experience model produce identical Gini values of 0.105, while the experience-only model performs substantially worse, with a Gini value of 0.021 (Fig. 3). This indicates that the experience variable provides little useful signal when claims are primarily driven by stochastic variation. The lift charts show a consistent pattern (Fig. 4): the actual loss cost relativity, the basic model prediction, and the combined model prediction closely track each other across the bins, while the experience-only model remains nearly flat with a lift score of 0.044. This suggests that historical experience alone has minimal ability to differentiate risk levels in this setting.

In the latent persistent risk scenario, model performance improves noticeably when the experience variable is included.

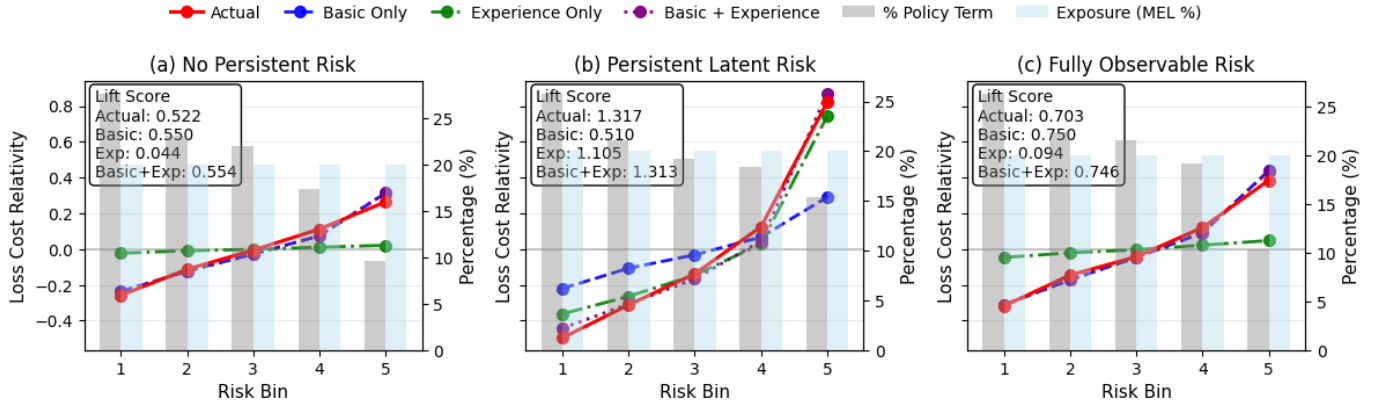


Fig. 4. Lift charts comparing predicted and observed loss cost relativities for the three risk scenarios.

The experience-only model achieves a Gini value of 0.249, significantly higher than the basic model's value of 0.149, while the combined model achieves the highest Gini value of 0.262 (Fig. 3). This represents an increase of approximately 76% in Gini relative to the basic model when the experience variable is incorporated, indicating substantially stronger risk segmentation. The lift charts confirm the same pattern observed in the Gini results (Fig. 4): the experience-only model closely follows the observed loss cost relativities, achieving a lift score of 1.105 compared with 1.317 for the actual outcome, whereas the basic model achieves only 0.510. The combined model achieves the highest lift score (1.313), but this value is only about 16% higher than that of the experience-only model. These results indicate that the experience variable effectively captures persistent hidden risk characteristics embedded in historical claims that are not explained by observable rating variables.

In the fully observable risk scenario, the addition of the experience variable provides little improvement in predictive performance. The basic model achieves a Gini value of 0.141, while the experience-only model performs poorly with a Gini value of 0.038. Adding the experience variable to the basic model leaves the Gini essentially unchanged at 0.140 (Fig. 3). The lift charts show a similar result (Fig. 4): the actual, basic, and combined model curves closely track each other across the bins, while the experience-only model remains nearly flat with a lift score of 0.094. Notably, the combined model lift (0.746) is slightly lower than the basic model lift (0.750), suggesting that the experience variable introduces minor noise rather than improving predictive performance when risk is already well explained by observable rating variables.

VI. CONCLUSION

This study provides a comprehensive analysis of the predictive role of experience variables in insurance pricing using a controlled simulation framework. By constructing three distinct risk environments, random risk, persistent latent risk, and fully observable risk, the analysis isolates the conditions under which historical claims experience contributes meaningful pre-

dictive information. The results show that experience variables provide substantial predictive value when persistent latent risk exists, as historical claims information captures underlying risk components that are not represented by observable rating variables. In contrast, when risk is primarily random or fully explained by observable predictors, the predictive contribution of experience variables becomes limited and provides little additional predictive benefit. These findings help clarify the role of experience-based features in insurance pricing models.

Future research could examine whether assigning greater weight to more recent claims experience improves the predictive value of experience variables, compared with approaches that treat historical claims information uniformly across time. Future work could also investigate interactions between experience variables and observable rating variables, which may provide further insight into how historical claims information complements traditional rating factors in risk modeling.

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